Brief history of MRI

1946 – Bloch and Purcell independently describe the NMR phenomenon

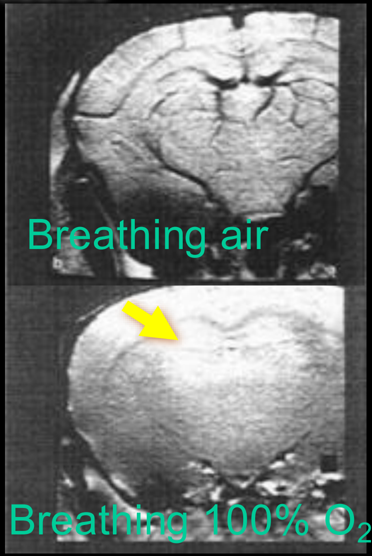
1952 – Bloch and Purcell – Nobel Prize in Physics

1971 – Damadian: NMR used to distinguish healthy and malignant tissues

1973 – Lauterbur: Back-projection MRI

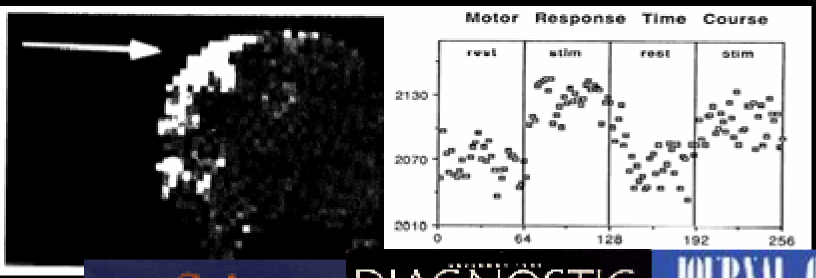
1975 - Ernst: Fourier Transform based MRI

1977 – Mansfield: Echo-Planar Imaging



1990 – Ogawa: BOLD and fMRI

1991 - Ernst – Nobel Prize in Chemistry



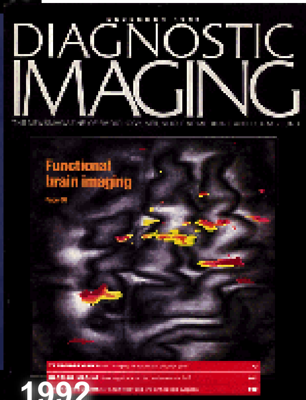
1991-1995: first task-related BOLD responses

Early '90s -fMRI-related analysis software development

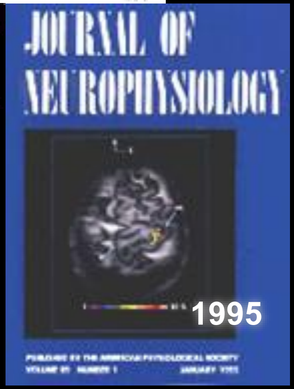
2003 – Lauterbur and Mansfield – Nobel Prize in Medicine

High-field (>7T scanner) applications

1991

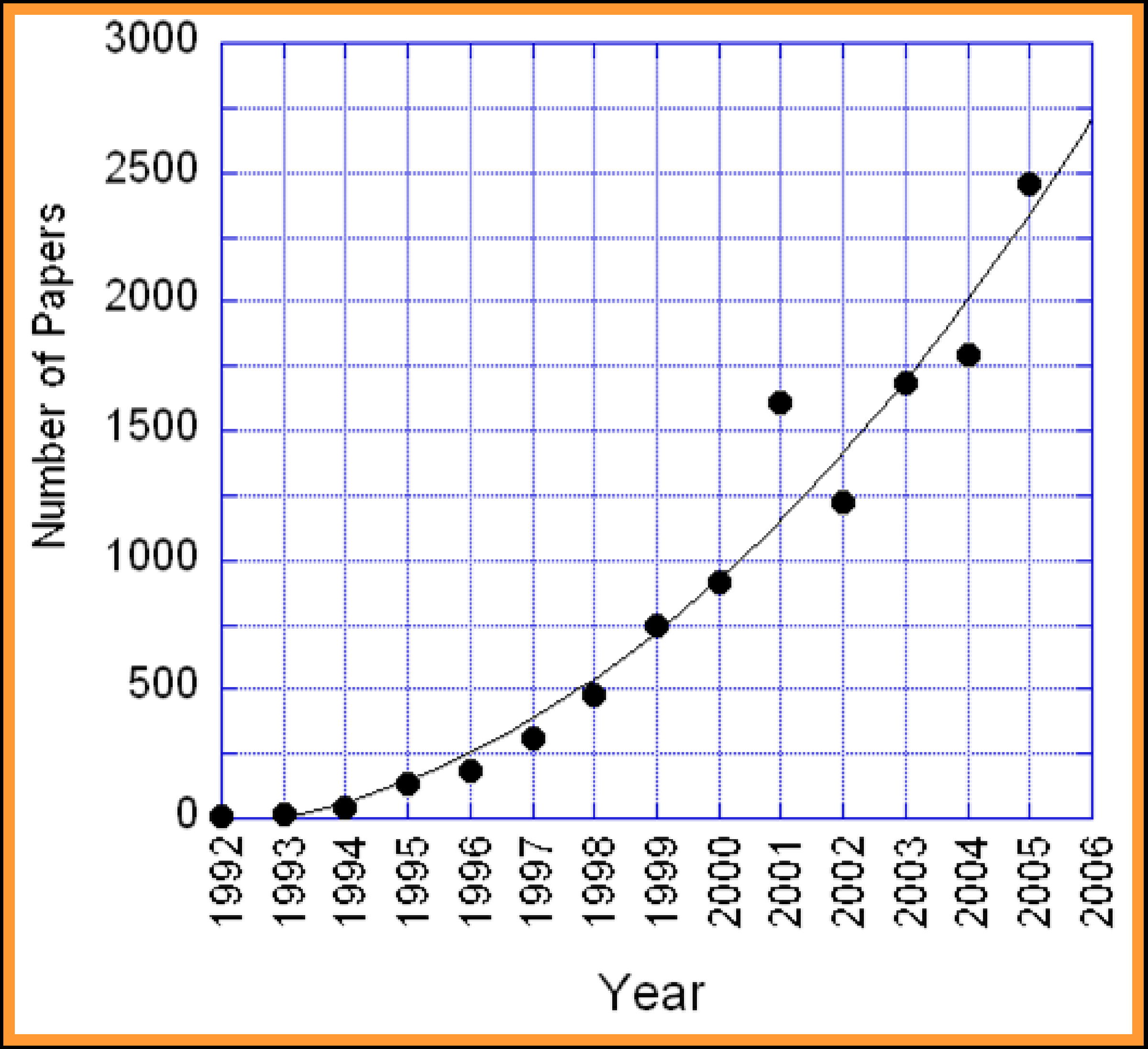


1992



1995

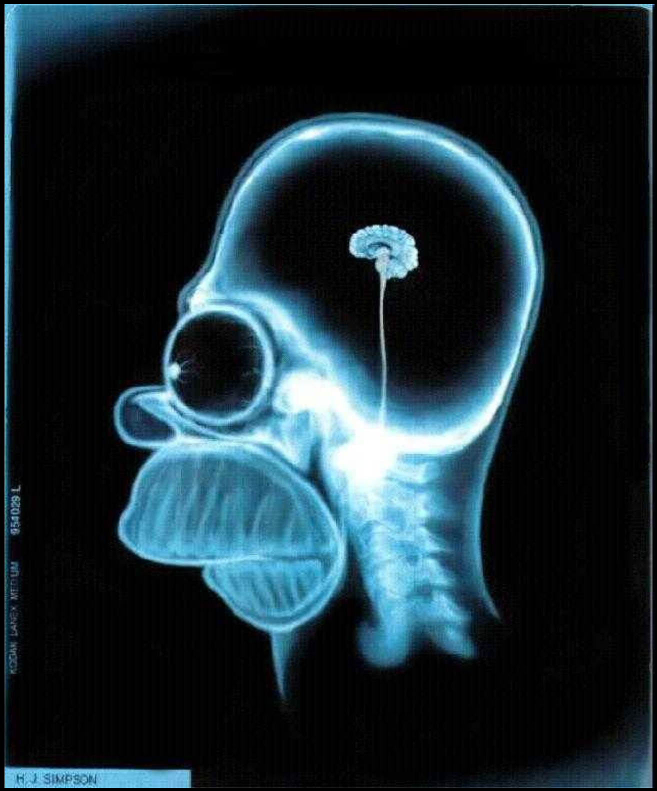
fMRI Papers Published per Year



Courtesy of Peter Bandettini

MRI vs. fMRI

MRI studies brain anatomy.



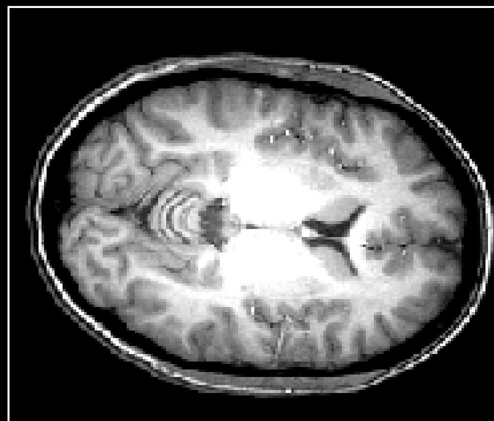
Functional MRI (fMRI) studies brain function.

Cosa otteniamo?

MRI vs. fMRI

MRI

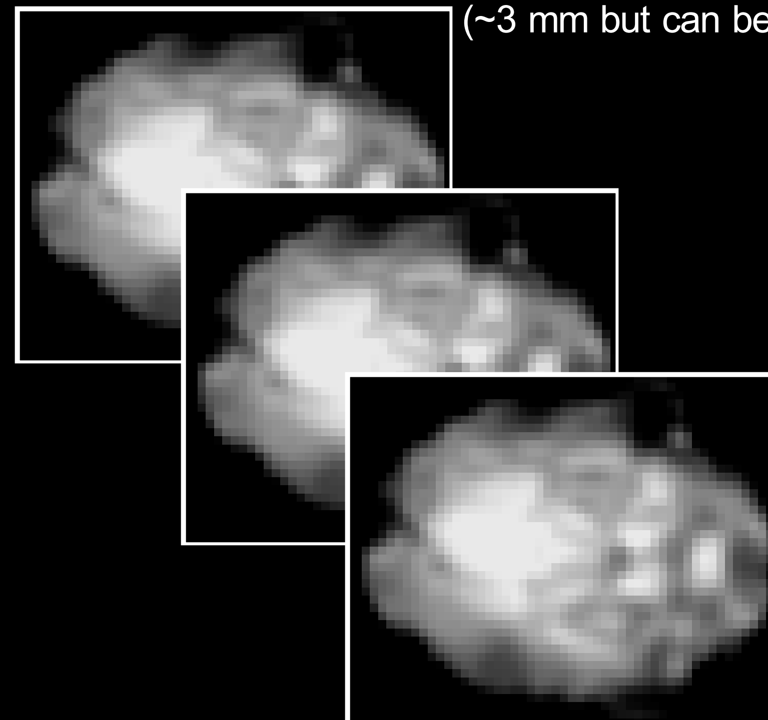
high resolution
(1 mm)



one image

fMRI

low resolution
(~3 mm but can be better)



. . .

many images
(e.g., every 2 sec for 5 mins)

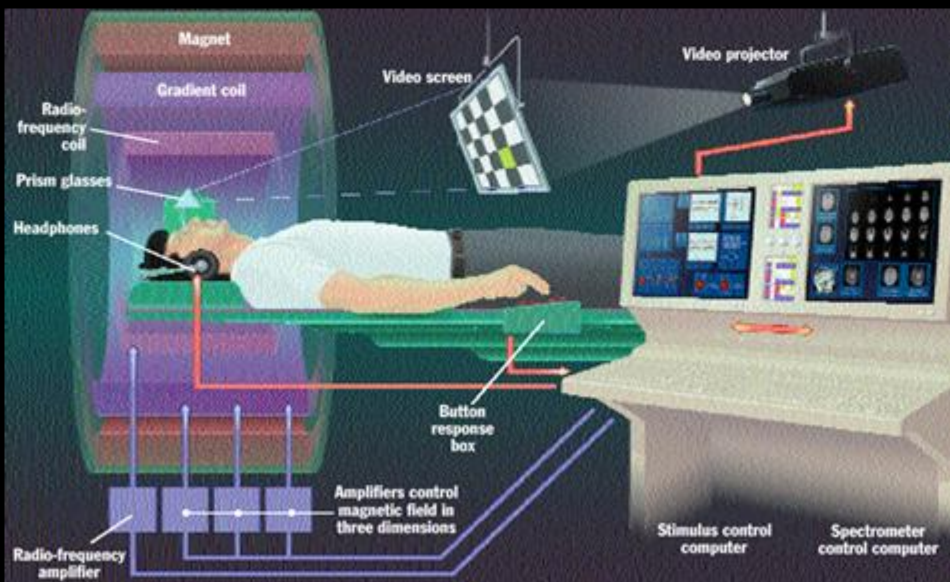
fMRI

Blood Oxygenation Level Dependent (BOLD) signal
indirect measure of neural activity

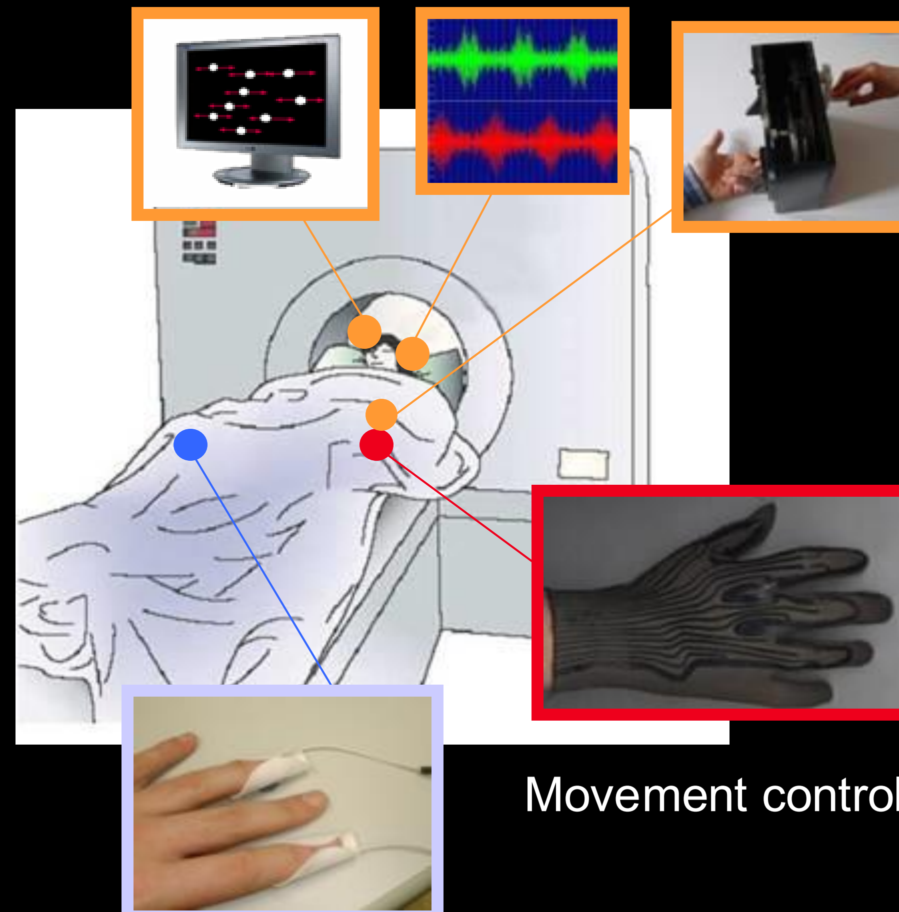
↑ neural activity ➔ ↑ blood oxygen ➔ ↑ fMRI signal

fMRI for Dummies

fMRI experimental setup



Visual, tactile and aural stimulation with MR-compatible devices

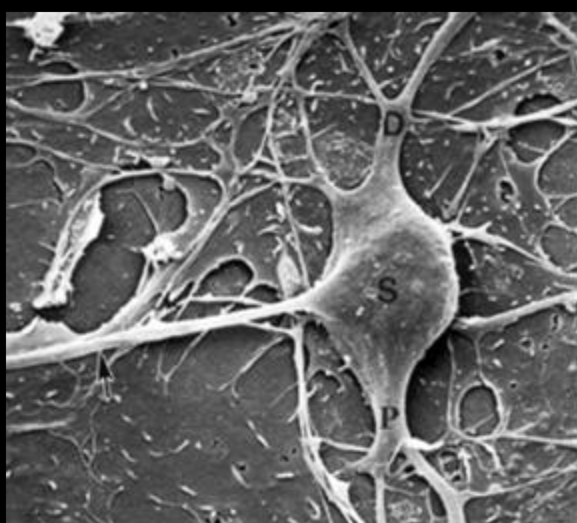


Movement control

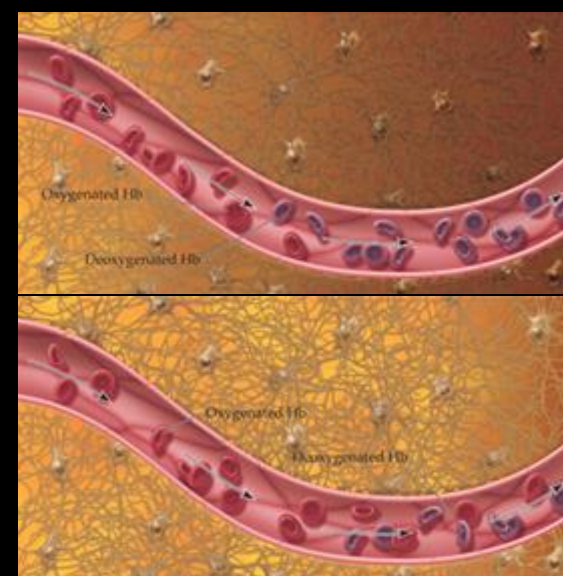
Measuring skin conductance and peripheral parameters (e.g. EKG, pO2)

In vivo brain functional methodologies

$\uparrow$ Neuronal synaptic activity $\rightarrow$ $\uparrow$ Activity of Na^+/K^+ pump $\rightarrow$ $\uparrow$ ATP request $\rightarrow$ $\uparrow$ Glucose and oxygen request $\rightarrow$ $\uparrow$ Cerebral blood flow supply $\rightarrow$ $\uparrow$ Glucose oxydative metabolism and ATP production

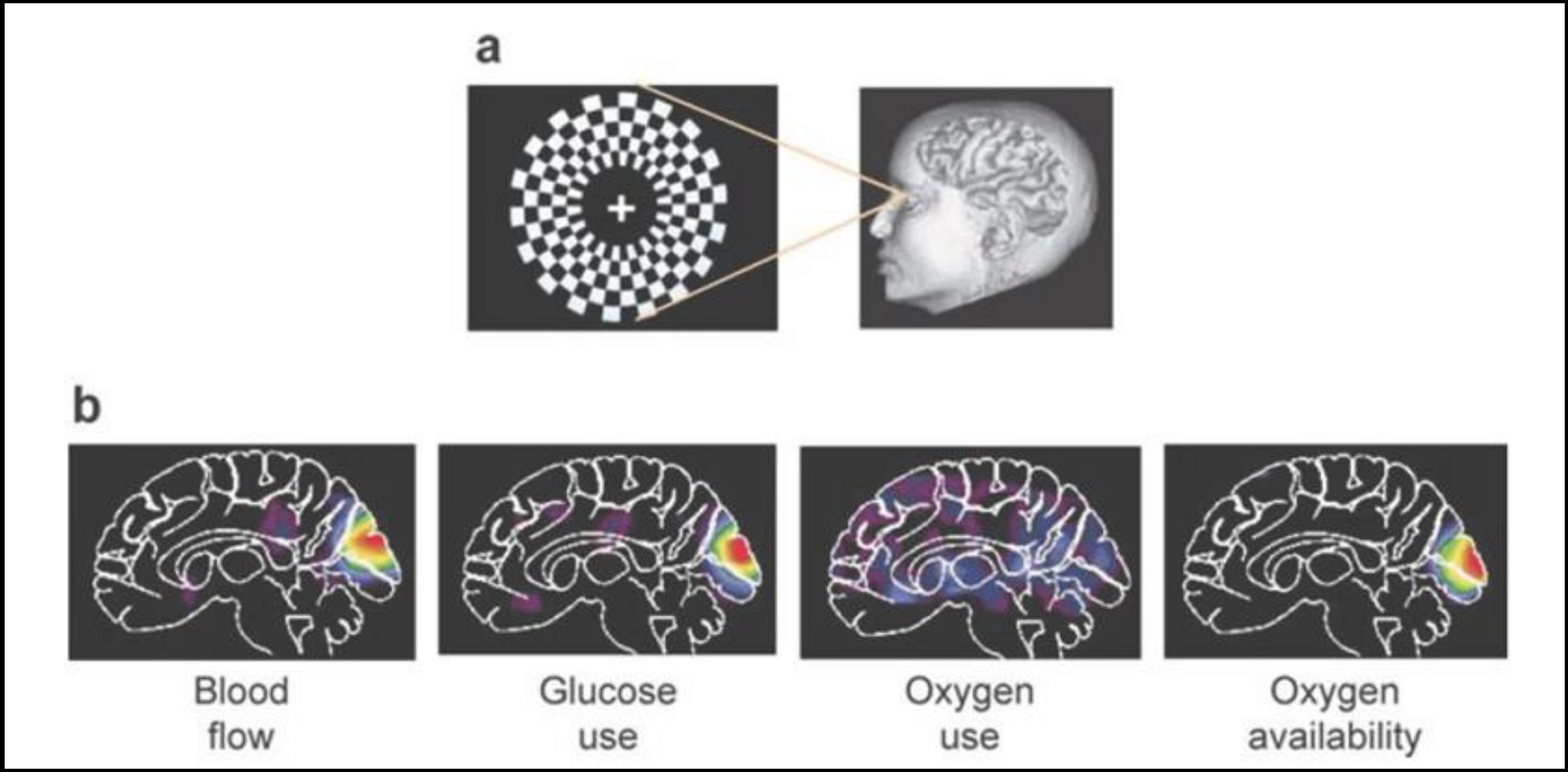


Neurovascular coupling



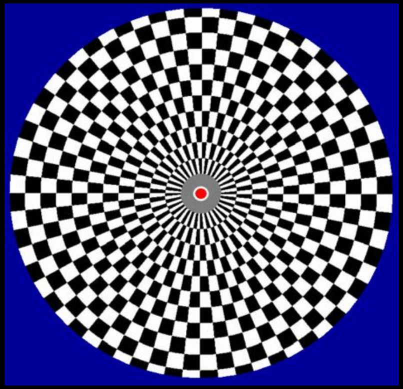
BOLD Contrast Imaging

What does happen to oxygen use?



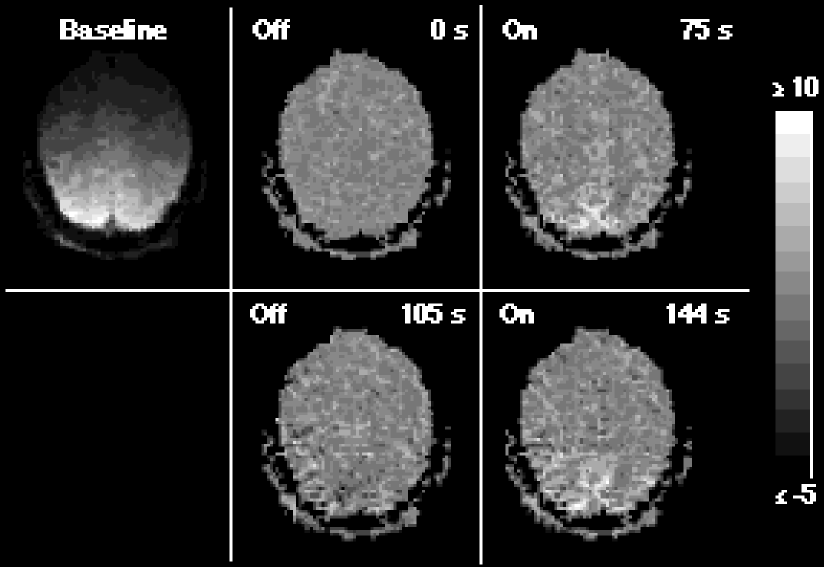
BOLD: Blood Oxygenation Level Dependent

fMRI Activation

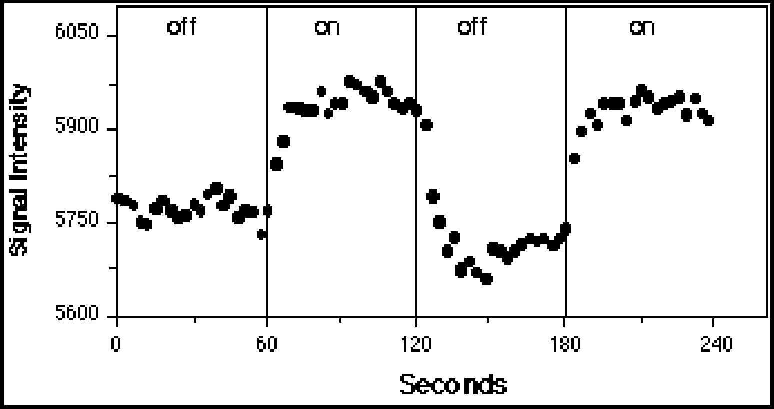


Flickering Checkerboard

OFF (60 s) - ON (60 s) -OFF (60 s) - ON (60 s) - OFF (60 s)



Brain Activity



Kwong et al., 1992

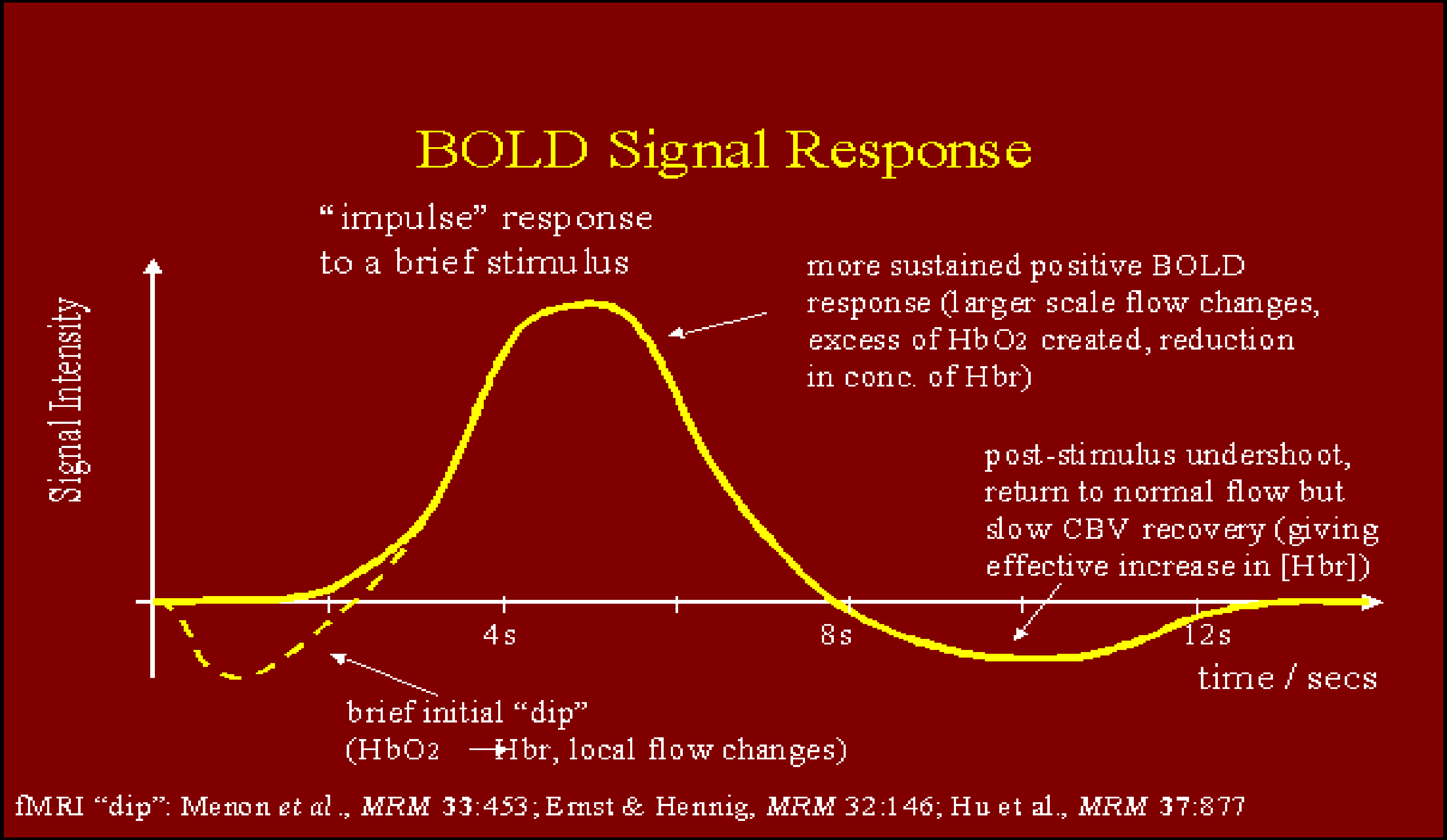
Time ⇨

fMRI for Dummies

BOLD Contrast Imaging

↑ deoxyhemoglobin ⇒ ↓ T_2^* ⇒ ↓ MR signal

↑ blood flow ⇒ ↓ deoxyhemoglobin ⇒ ↑ T_2^* ⇒ ↑ MR signal

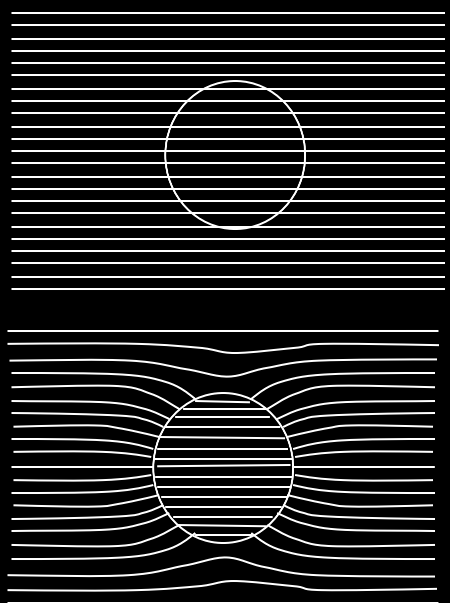


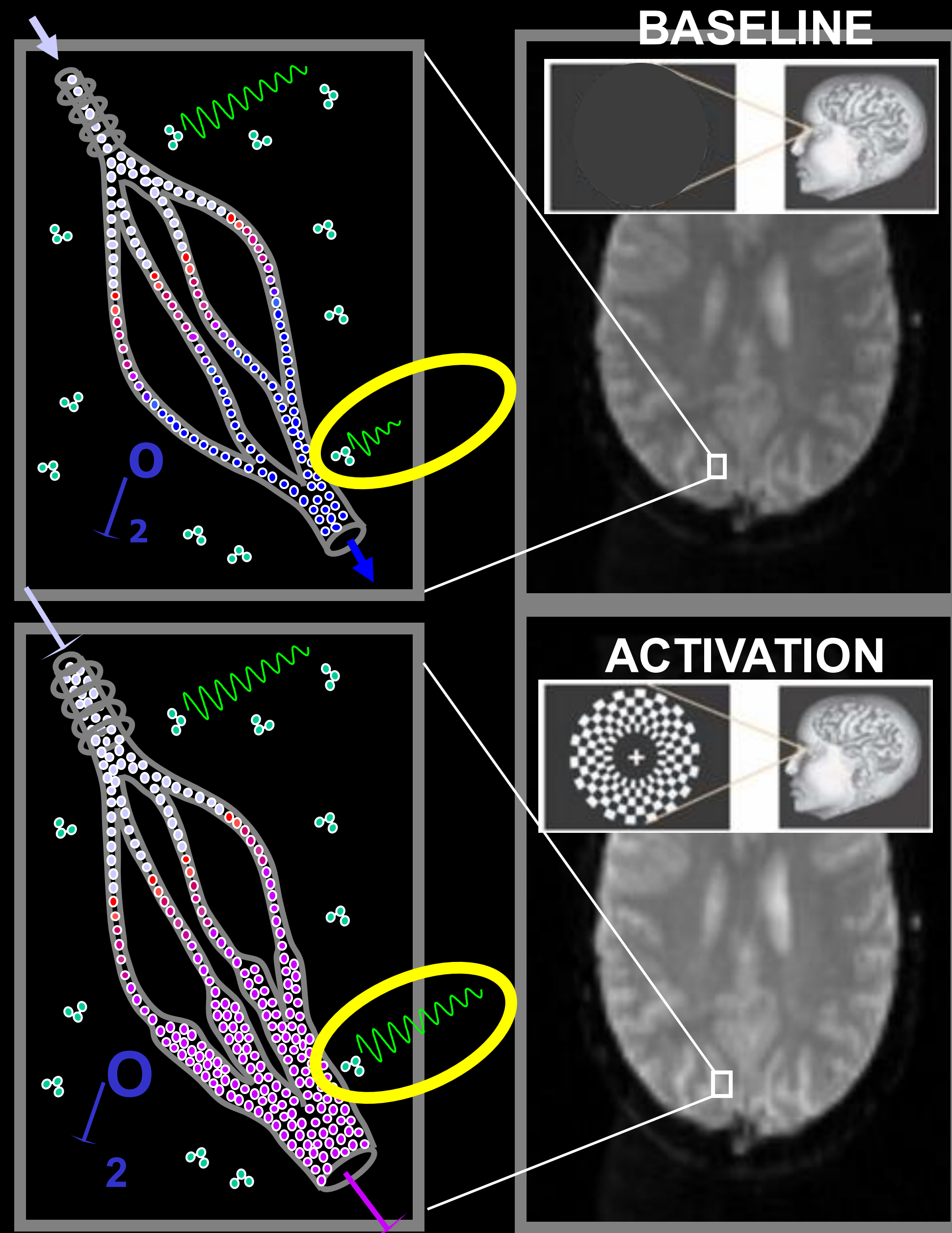
BOLD Contrast Imaging

Oxygenated and deoxygenated hemoglobin have different magnetic properties

oxyHb

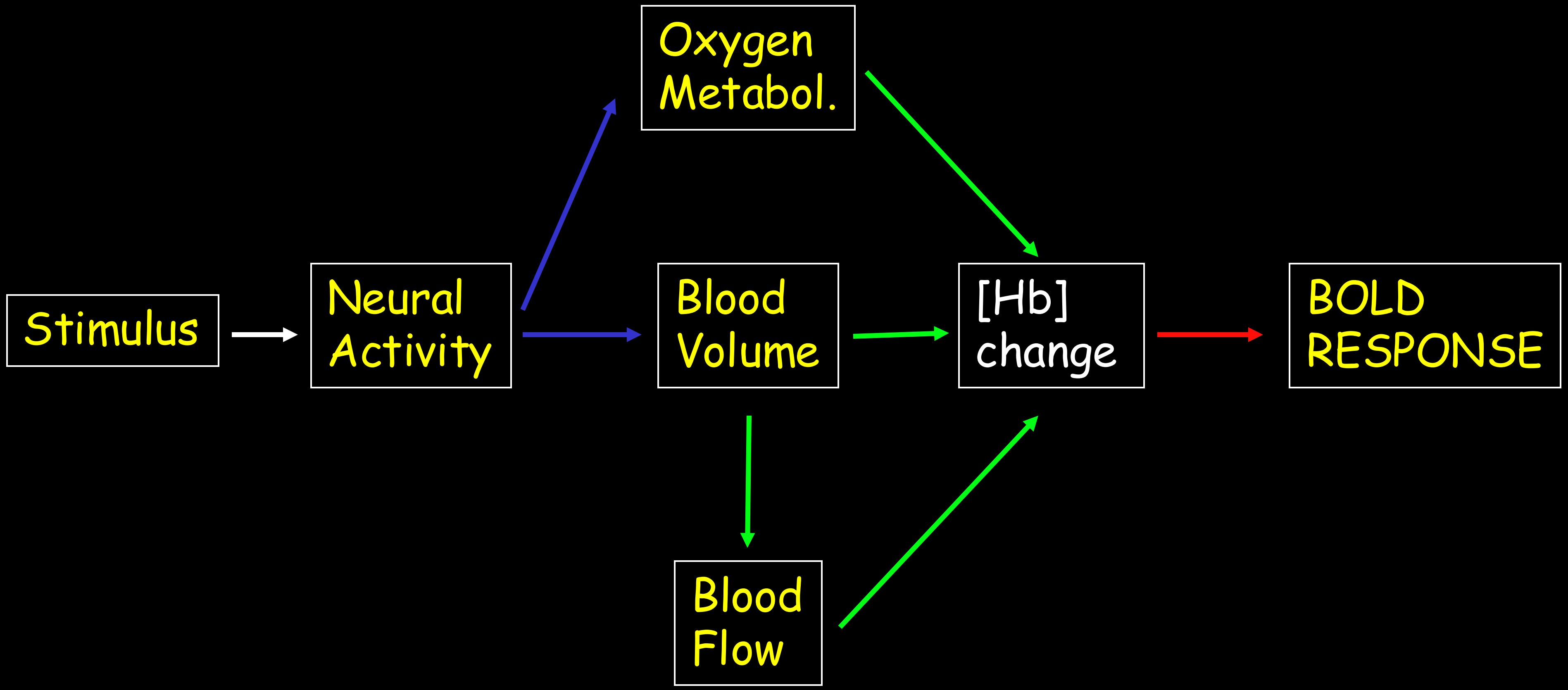
deoxyHb





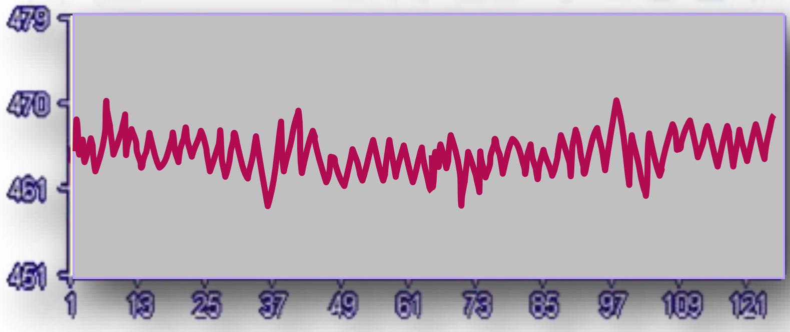
L. Pauling, C. D. Coryell, *PNAS*, 1936.
K.R. Thulborn, J. C. Waterton, et al., *Biochim. Biophys. Acta.*, 1982
S. Ogawa, T. M. Lee, A. R. Kay, D. W. Tank, *PNAS*, 1990.

BOLD Contrast Imaging



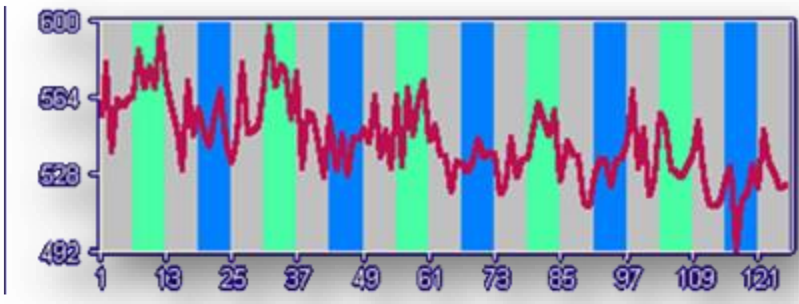
IMT SCUOLA ALTI STUDI LUCCA

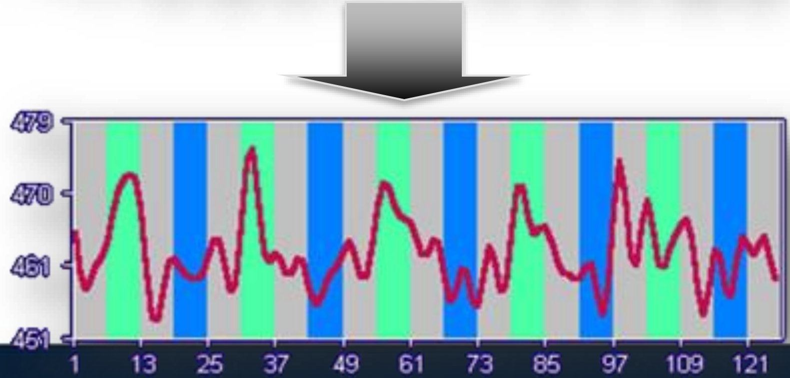
Le acquisizioni del segnale BOLD

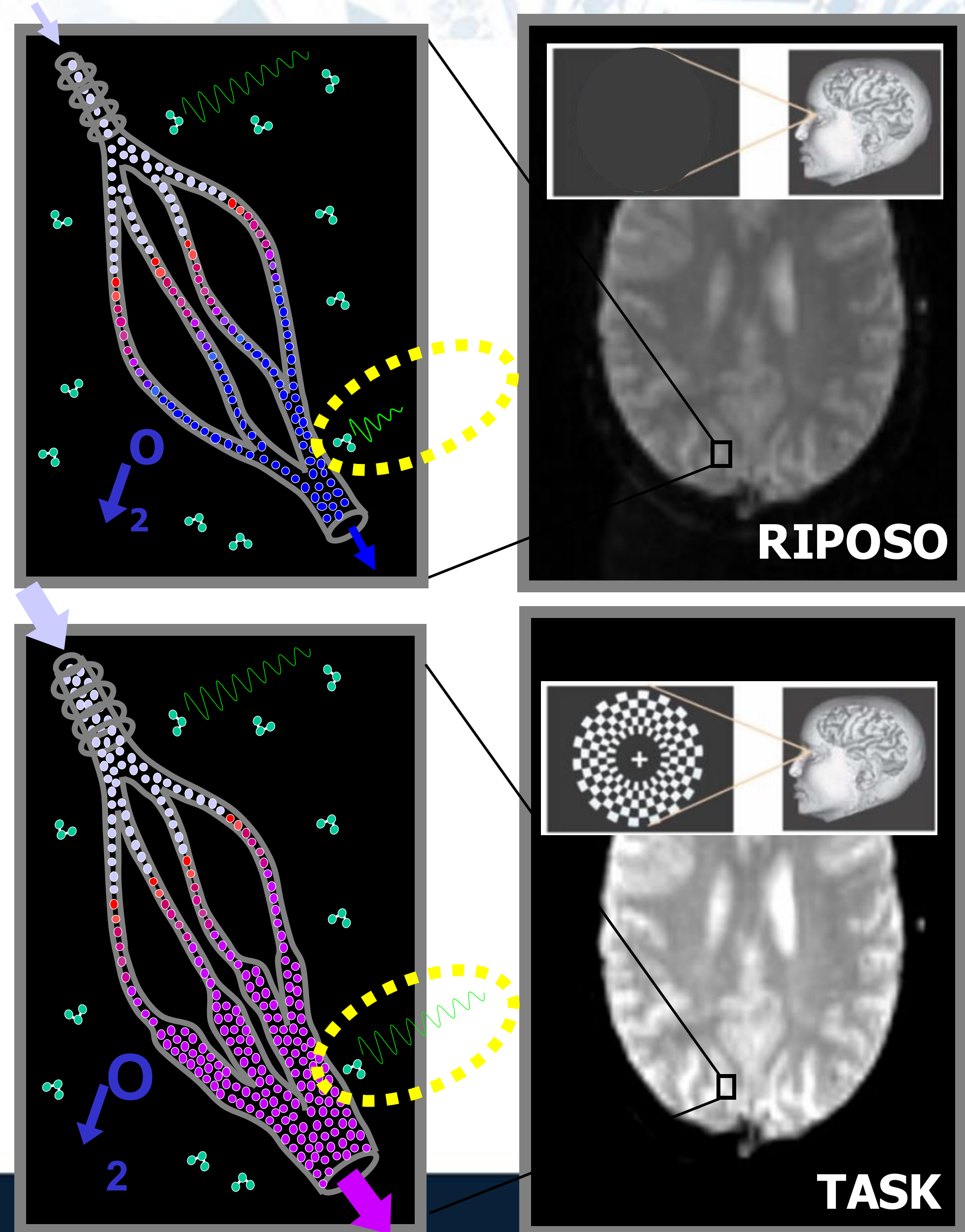


STIMOLI ON – TASK

STIMOLI OFF - RIPOSO



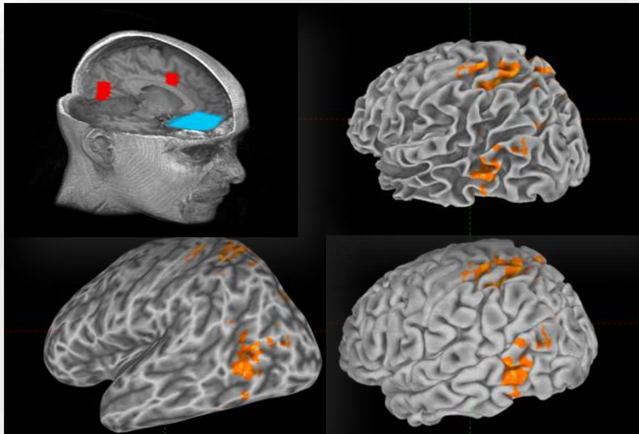




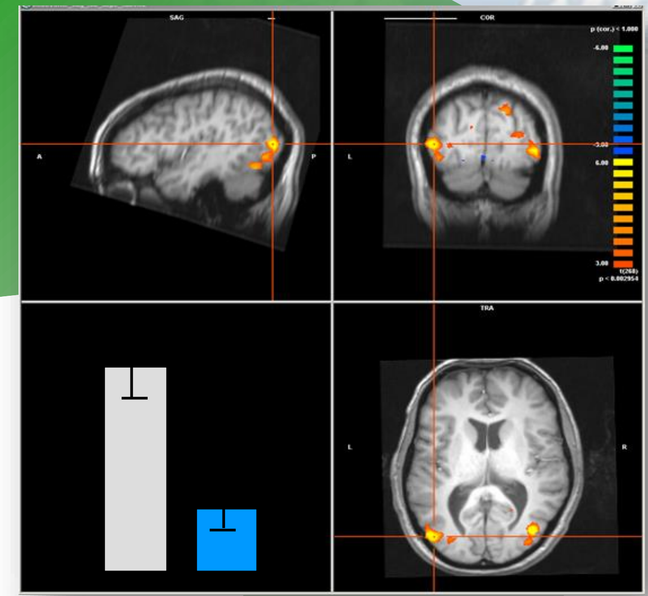
IMT SCUOLA ALTI STUDI LUCCA

Le analisi dei dati funzionali

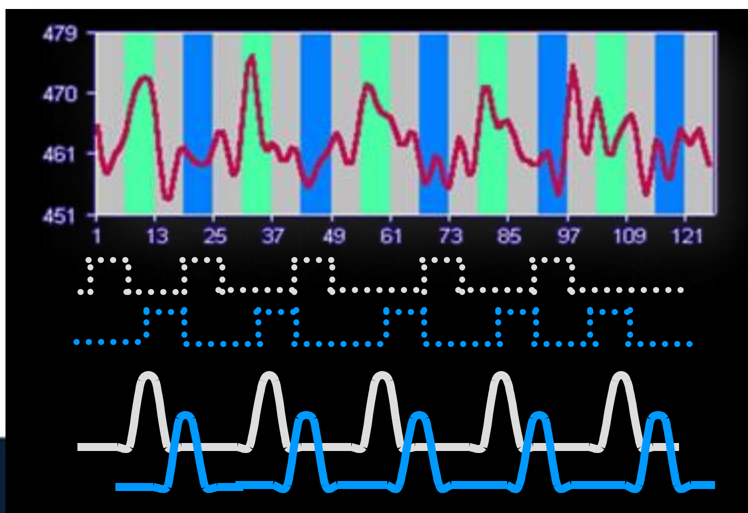
Visualizzazione di superfici corticali con le relative attivazioni

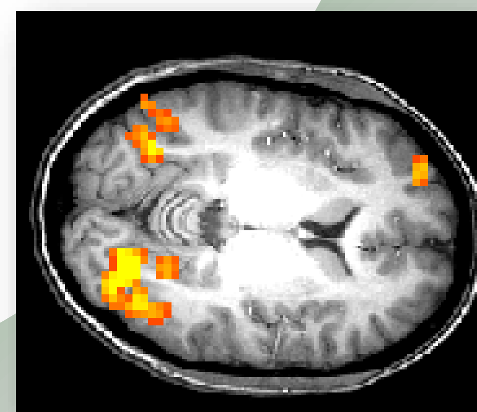


Passaggio 2D→3D Analisi quantitativa nelle regioni d' interesse



Correlazione con modello di risposta emodinamica





L'attività correlata al modello è rappresentata come un'attivazione nei singoli voxel

Functional Magnetic Resonance Imaging

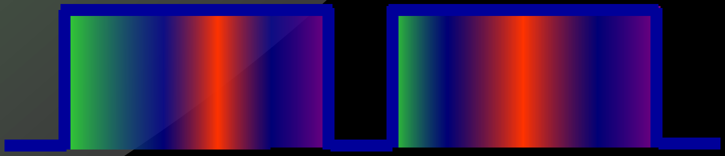
- **Basic physical principles**
- **BOLD signal as an indirect measure of neuronal activity**
- **Basics of data analyses**

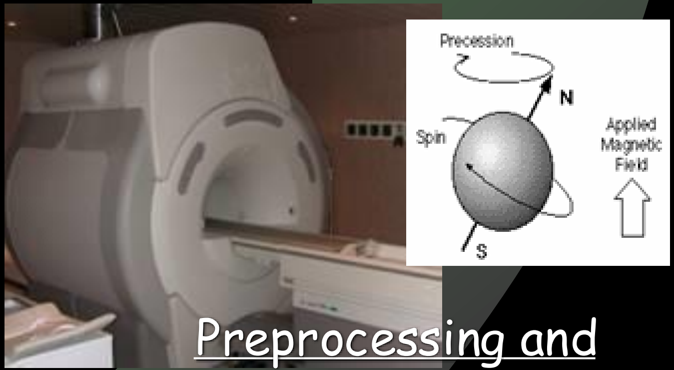
fMRI experimental flow chart

Rationale: define your hypothesis

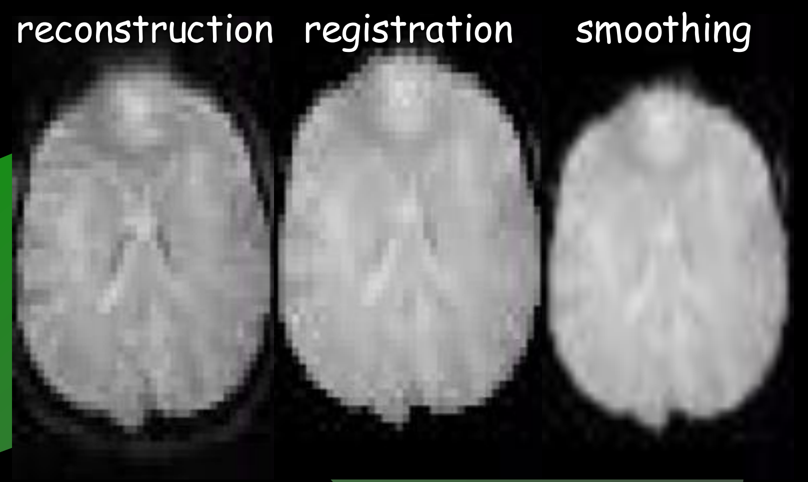
Subject selection and screening

Experimental paradigm

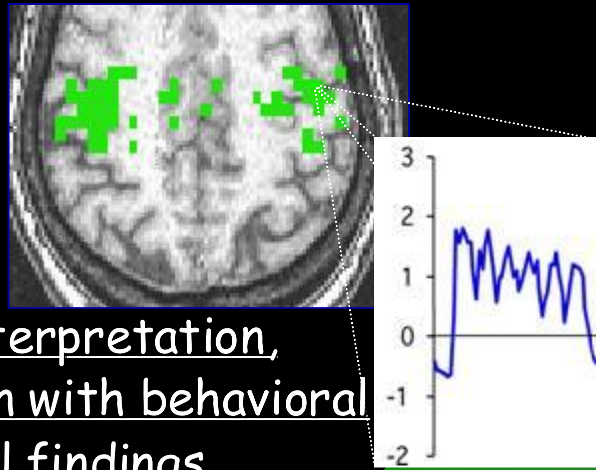




Preprocessing and experimental session



Data analyses (postprocessing)

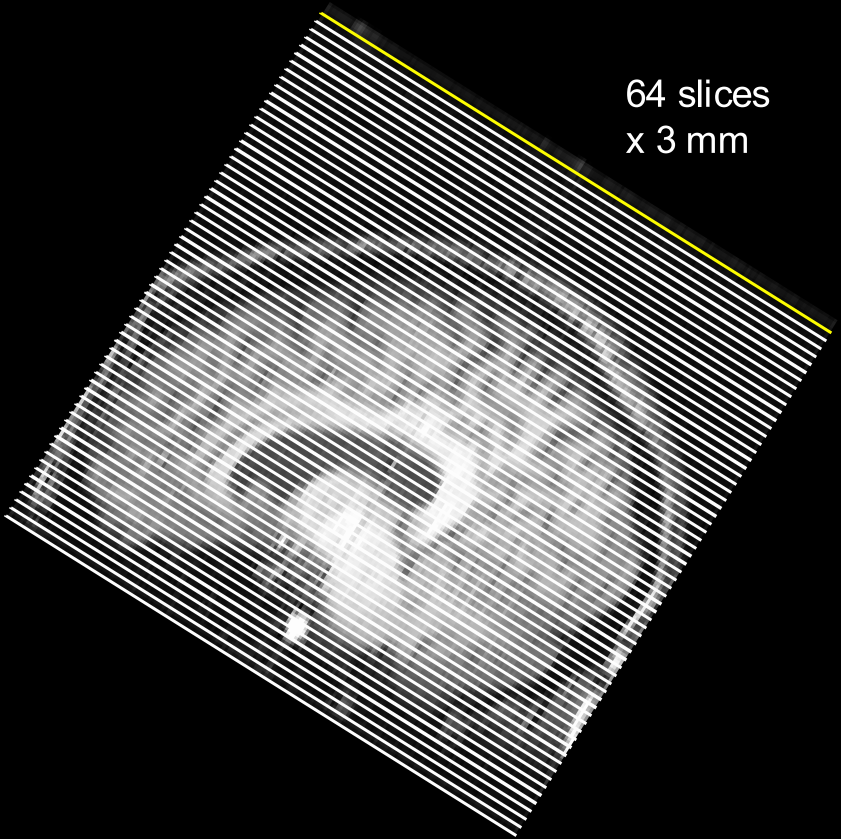


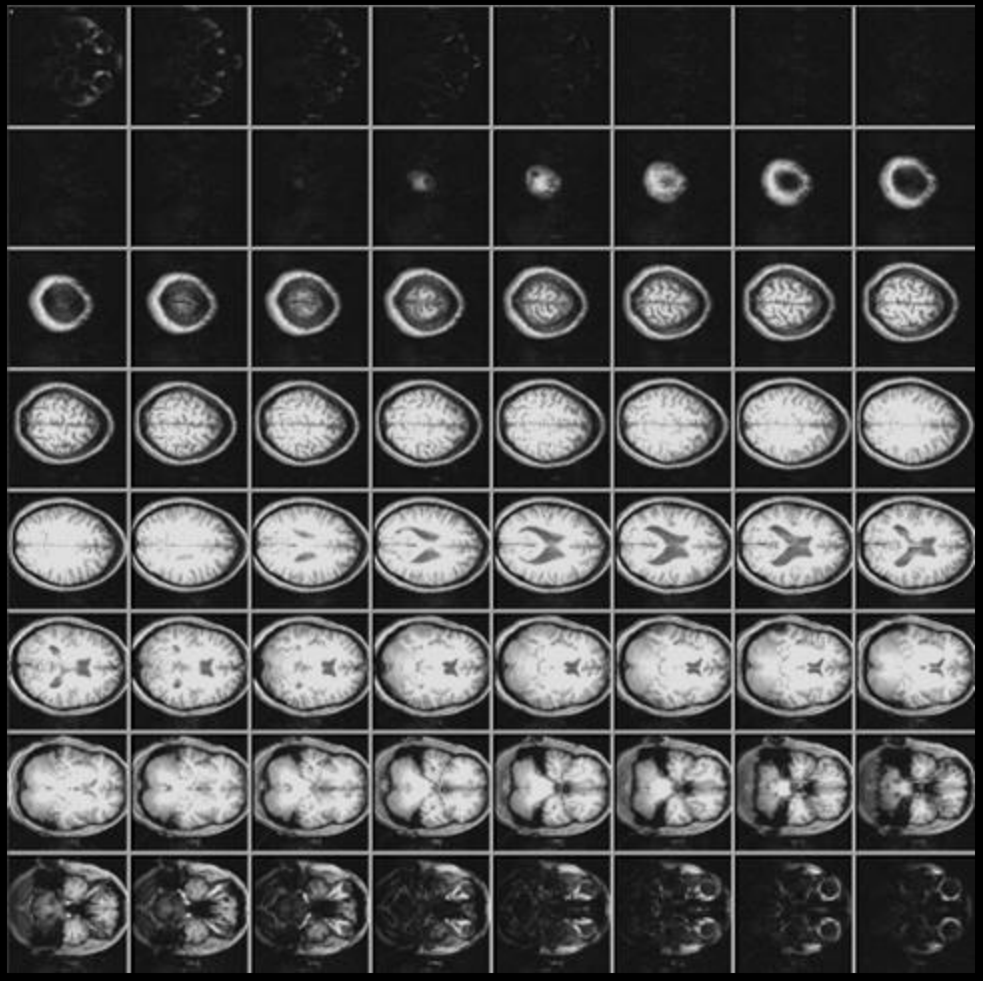
Results interpretation, correlation with behavioral and clinical findings

fMRI Experiment Stages: Anatomicals

4) Take anatomical (T1) images

- high-resolution images (e.g., 0.75 x 0.75 x 3.0 mm)
- 3D data: 3 spatial dimensions, sampled at one point in time
- 64 anatomical slices takes ~4 minutes





fMRI for Dummies

Recipe for MRI

1) Put subject in big magnetic field (leave him there)

2) Transmit radio waves into subject [about 3 ms]

3) Turn off radio wave transmitter

4) Receive radio waves re-transmitted by subject
   - Manipulate re-transmission with magnetic fields during this *readout* interval [10-100 ms: MRI is not a snapshot]

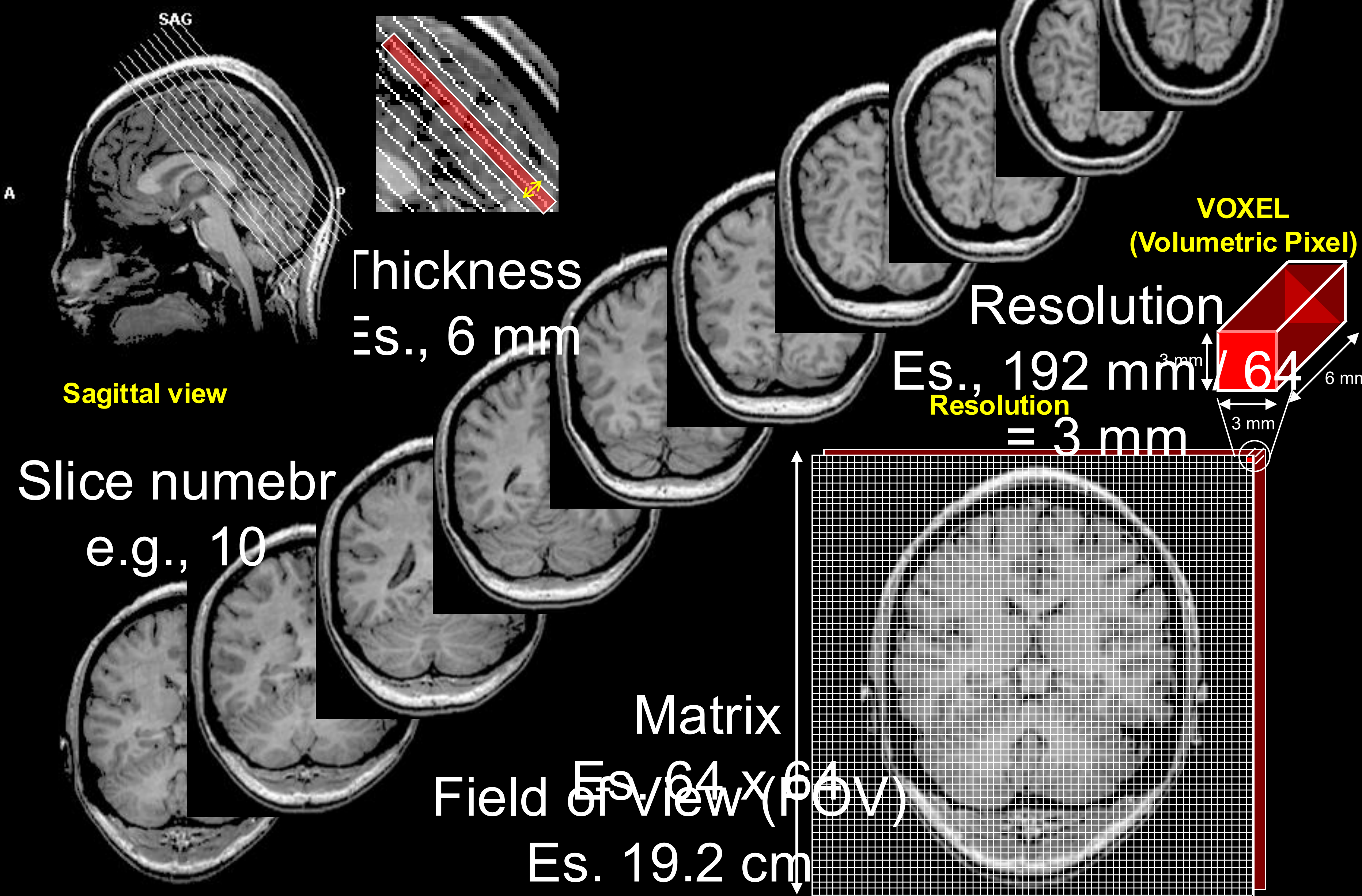
5) Store measured radio wave data vs. time
   - Now go back to 2) to get some more data

6) Process raw data to reconstruct images

7) Allow subject to leave scanner (this is optional)

Source: *Robert Cox's web slides*

Data acquisition



Different strategies to neural recruitment

1. Block Design
2. Mixed Block Design
3. Slow Event-Related
4. Fast Event-Related
5. Self-Paced
6. Sparse intermixed